

Jan Bauer

Homepage: japhba.github.io

EDUCATION

- Ph.D. student** in theoretical neuroscience 10/2022–
Gatsby Unit, University College London, United Kingdom
Edmond and Lily Safra Center, The Hebrew University of Jerusalem, Israel
Working on kernel descriptions of chaos in recurrent models of the brain.
Advisors: Agostina Palmigiano and Jonathan Kadmon
- Research stay** 11/2023–04/2024
Summerfield lab, University of Oxford, United Kingdom
Modelled cognitive flexibility with linear neural networks.
- M.Sc. in physics** GPA 1.0/1.0, with distinction 2019–2022
RWTH Aachen University, Germany
Focussed coursework on Statistical Mechanics of Neural Networks, Theoretical Neuroscience, Deep Learning in Physics Research, Quantum Information, and Computational Physics
- Master's thesis** 09/2021–08/2022
Juelich Research Center, Germany
Analysis of computation in cortical networks by Gaussian process regression.
Advisors: Prof. Moritz Helias, Dr. Christian Keup (now EPFL Switzerland)
- Bilateral graduate exchange program** 09/2019–03/2020
The University of Tokyo, Japan
Focussed coursework on Tensor Networks and Universal Biology, which investigates stem cell differentiation and evolution by a mathematical, complex systems approach. Member of Japanese university choir “The White Rose”.
- B.Sc. in physics** GPA 1.3/1.0, with distinction 10/2016–09/2019
RWTH Aachen University, Germany
- Bachelor's thesis** 03/2019–09/2019
Juelich Research Center, Germany
Description of unsupervised learning in Boltzmann machines via Feynman diagrams.
Advisors: Prof. Moritz Helias, Dr. Tobias Kuehn (now ENS Paris)
- 4x teaching assistant in theoretical physics and lab courses, mentoring 1st to 10th semester students.*

PUBLICATIONS

- Bauer J.***, De Schamphelaere C.*, Karvonen A., Luick N., Nanda N., 2026. *Building Better Activation Oracles*. **ICML 2026 Mechanistic Interpretability Workshop** 📄
- De Schamphelaere C.*, **Bauer J.**, Nanda N., Ong E., 2026. *LoRAcles: Self-Supervised Weight-Space Interpretability at Scale*. **NeurIPS 2026 (Oral)** 📄
- Bauer J.**, Keup C., Kadmon J., Helias M., 2026. *Discrete signaling mediates chaotic regularization in recurrent neural networks*. **arXiv preprint** 📄

Bauer J., Fischer K., Helias M., Palmigiano A., 2026. *A unified theory of feature learning in RNNs and DNNs*. **ICML 2026** 

Schutzeichel L., **Bauer J.**, Bouss P., Musall S., Dahmen D., Helias M., 2025. *Recurrent dynamics underlying transient neural representations*. **bioRxiv** 

Schwarcz J.*, **Bauer J.***, Rajabi, H., Marmur, G., Lottem, E., Kadmon, J., 2025. *Neural mechanisms of flexible perceptual inference*. **PLOS Comp. Biology** 

Rubruck J., **Bauer J.**, Saxe A., Summerfield C. *Neural mechanisms of flexible perceptual inference*. **CCN 2025 Conference proceedings**

Sandbrink K.* **Bauer J.***, Proca A.M.*, Saxe A., Summerfield C., Hummos A.* , 2024. *Flexible task abstractions emerge in linear networks with fast and bounded units*. **NeurIPS 2024 (Spotlight)** 

PROFESSIONAL EXPERIENCE

MATS 10.0 Fellowship Berkeley, United States <i>Built meta-models (AOs, LoRAcles) to audit LLMs when black-box monitors fail.</i> Mentor: Neel Nanda	02/2026–08/2026
PIBSS Fellowship London Initiative for AI Safety, London, United Kingdom Mechanistic interpretability of neural geometry behind chain-of-thought, public talk Mentor: Jan Hendrik Kirchner (Anthropic)	06/2024–09/2024
Research Internship Bosch Center for Artificial Intelligence, Renningen, Germany <i>Co-developed and implemented architectural extensions of Bayesian version of the Neural Process model.</i> Advisor: Michael Volpp	03/2021–06/2021

HONOURS

HUJI International PhD Talent Scholarship	since 2023
Springorum memorial coin for outstanding graduates	2022
Scholarship fellow of the Konrad Adenauer Foundation (one of 13 academic talent promoters established by the Federal Ministry of Education and Research)	2016-2022
Fellow of RWTH Aachen University's Dean's List of top 5% students	since 2019
Valedictorian in Abitur A-Levels	2016

PRESENTATIONS

Invited talk Temporal feature learning , Institute for Advanced Simulation, Juelich, Germany	10/2025
Poster Recurrent networks under constraint of sparse reward learn interacting belief state dynamics , COSYNE 2024, Lisbon, Portugal	03/2024
Talk Can discrete neurons be useful? , ELSC Annual Retreat, Nahsholim, Israel	06/2023
Poster Discrete communication mediates effective regularization in chaotic recurrent networks , COSYNE 2023, Montréal, Canada	03/2023
Random Matrix Theory for Machine Learning , Parallel Sessions of INM-6 Annual Retreat	05/2022
Inference with Graphical Models , Book Club of Institute for Neuroscience and Medicine, Juelich Research Center	10/2021
Proof-read and advised high school textbook together with teacher Dr. Bardo Diehl at didactics congress MNU Aachen in front of 40 participants (<i>"Zentrale Experimente für das Abitur", Cornelsen 2017</i>)	04/2017

SCHOOLS & WORKSHOPS

12-day workshop "Analytical Connectionism" University College London, United Kingdom	09/2023
3-week summer school "Mathematical Methods in Computational Neuroscience" NTNU, Norway	07/2023
5-day workshop "Recent advances in understanding artificial and biological neural networks" Les Houches School of Physics, France	02/2023
5-day summer school "Reinforcement Learning" Vrije Universiteit Amsterdam	06/2022

TEACHING EXPERIENCE

Statistical Physics and Field Theory	summer 2022
Theory of Electromagnetism	winter 2022
Preparatory math course for computer scientists	10/2020
Theory of Thermodynamics	winter 2020
Introductory lab course for physicists	09/2018

LEADERSHIP AND OUTREACH

Social commitment

Volunteer at the City of Aachen, supporting Egyptian family with homework and administrative tasks 2017–2020

Co-organisation of "LernFair"-AI lessons, part of a project aimed at high school students during the pandemic winter 2021

Full-time scholarship by the Konrad Adenauer Foundation 2016–2022

Elected **spokesperson** of local group of 25 students 2019–2020

Increased participation of students by bundling proposals for engaging and meaningful group activities, such as a volunteer week in kindergarten in socially deprived suburb of Aachen

Initiated and organized **4-day seminar on the scientific voice in democracies**, with invited speakers on the philosophy of science, politics, recent societal challenges, and science communication 04/2022

[Service](#) to inform about cancellation of school lessons in the early morning before classes start 2015

Development of [mobile app](#) to create a precise elevation map of Aachen by use of barometer data of phones, targeted at finding a least elevation bicycle route 2016

[Application of machine learning and Fourier decomposition](#) to successfully remove chequered paper background from handwritten lecture notes 2020

SOFTWARE AND LANGUAGE PROFICIENCY

Software

Python, JAX, PyTorch

3D computer graphics with Blender

Languages

German (native)

English (academic proficiency, daily usage)

French (good, DELF A2)

Japanese (good, weekly practice with Tandem partner)

RECREATIONAL ACTIVITIES

Bicycle touring and medium distance running (olympic-distance triathlon in September 2022)

Singing in university choir (tenor voice)

REFERENCES

Dr. Agostina Palmigiano, Ph.D. advisor, Gatsby Unit, UCL

Dr. Jonathan Kadmon, The Hebrew University of Jerusalem

Prof. Moritz Helias, M.Sc. thesis advisor, Juelich Research Center

Dr. Michael Volpp, research internship advisor, Bosch Center for Artificial Intelligence

Dr. Christian Keup, M.Sc. thesis advisor, EPFL

Dr. Tobias Kuehn, B.Sc. thesis advisor, Ecole Normale Supérieure Paris

Updated: September 25, 2026